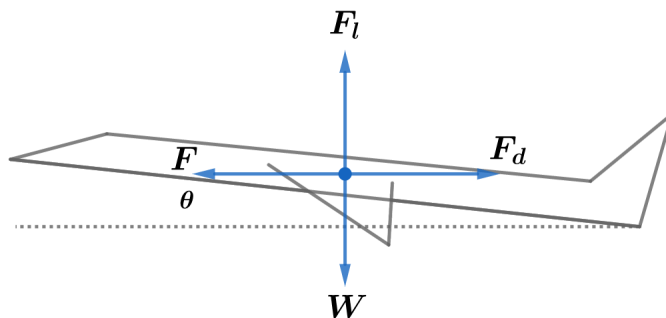


2018B F=ma Exam: Problem 16

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Given the relative velocity v of the plane to the air, we know

$$F_l = k_1 v^2 \theta$$

$$F_d = k_2 v^2$$

To fly at constant velocity, we have

$$W = F_l = k_1 v^2 \theta$$

$$F = F_d = k_2 v^2$$

Thus,

$$\theta = \frac{W}{k_1 v^2}$$

$$P = Fv = k_2 v^3$$

In a region with tailwind of velocity w , the relative velocity of the plane to the air becomes $v \rightarrow v - w$. Hence, θ must increase and P must decrease to maintain constant ground speed so the answer is C.